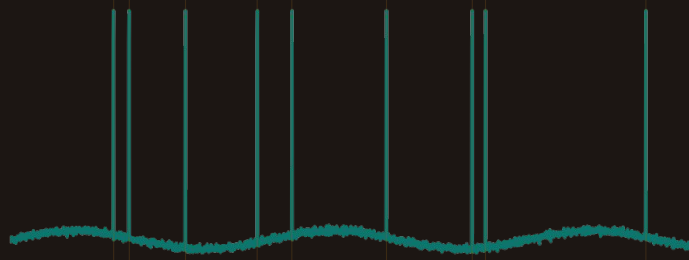


ACNEI - Mathematical Modeling for Brain Data(MAMBA) Workshop

# Calculus and Differential Equations for Neural Dynamics

*How calculus and differential equations describe neural behaviour that changes over time*



# Model Change ?

“All models are wrong but some are useful ”.  
George Box

## 1 Why

Understanding the behavior is about understanding their evolution in time, and that is “Change”.

## 2 What

A system that evolves over time under the influence of some factors.

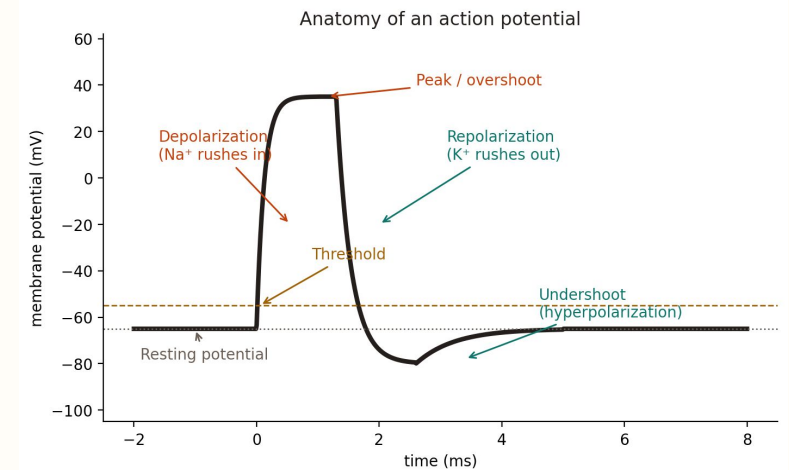
## 3 How

Calculus and Differential Equations

# 1

## Why change over time matters

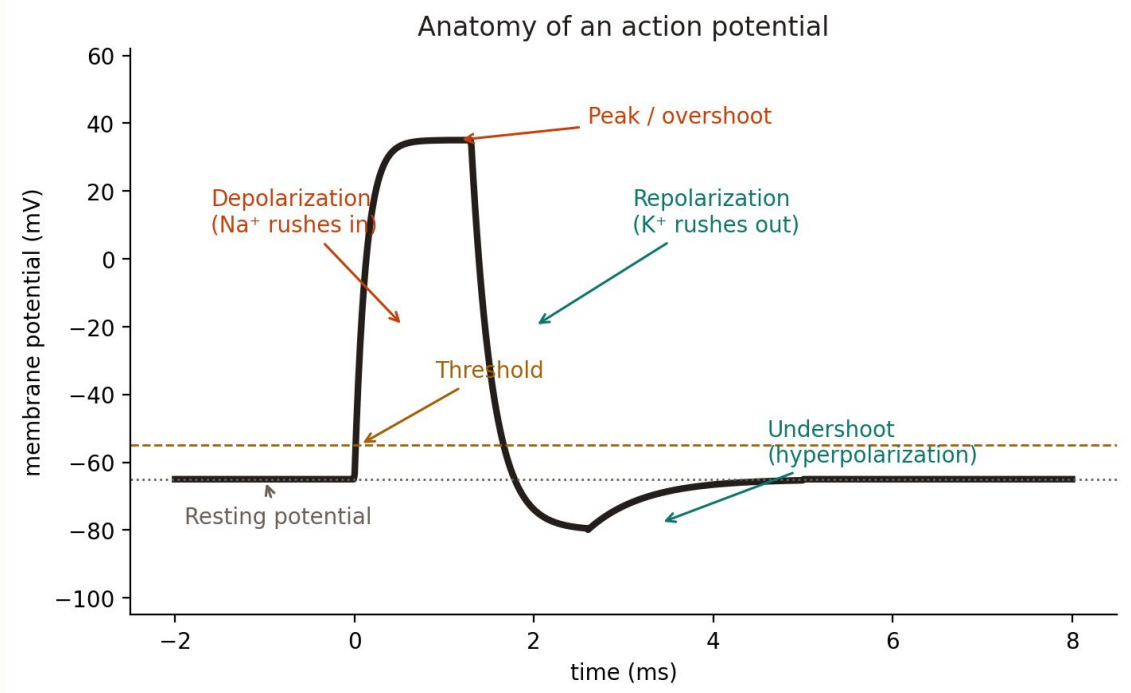
*A single number tells you almost nothing. The trajectory tells you everything.*



# Why it matters

## *Take membrane potential.*

A single reading of  $-58.3$  mV tells you almost nothing — is the neuron about to fire? Recovering from a spike? Resting? What actually matters is the trajectory: which way  $V$  is heading, and how fast. That's exactly what the action potential trace on the right shows — the same quantity, but followed through time instead of frozen at an instant.



*Membrane potential is NOT just a number. It's a trajectory with phases, and the interesting biology is in how it moves between them.*

# Why change over time matters

*“If you can describe, in words, how something changes, you already have the intuition for a differential equation.”*

Almost nothing interesting in the nervous system is static. A neurotransmitter concentration rises after release and falls as it's cleared. A population's firing rate ramps up during a decision and back down after. A synaptic weight strengthens slowly over hours of learning. In every case, **the interesting biology is encoded in HOW the quantity changes; which is exactly what calculus and differential equations describe.**

## Key idea:

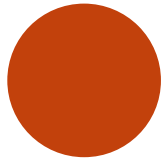
a model of a static snapshot tells you what a system looks like right now. A model of change over time tells you what it will do next ; which is usually the question we actually care about.

# Change happens at every scale



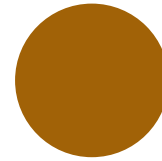
## Molecular

Channel gating variables evolve within milliseconds.



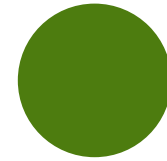
## Cellular

Membrane potential rises and falls with synaptic input.



## Circuit

Population firing rate evolves over tens of milliseconds.



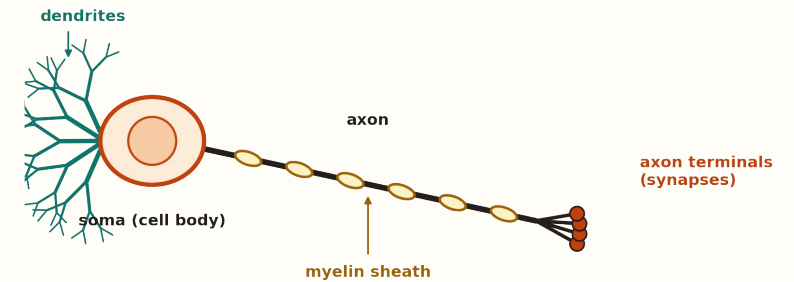
## Systems

Neuromodulator levels & synaptic weights evolve over seconds, minutes, or longer.

# 2

## Variables, parameters, and functions

*Get this distinction right, and every equation from here on gets easier to read.*



# Variable vs Parameter

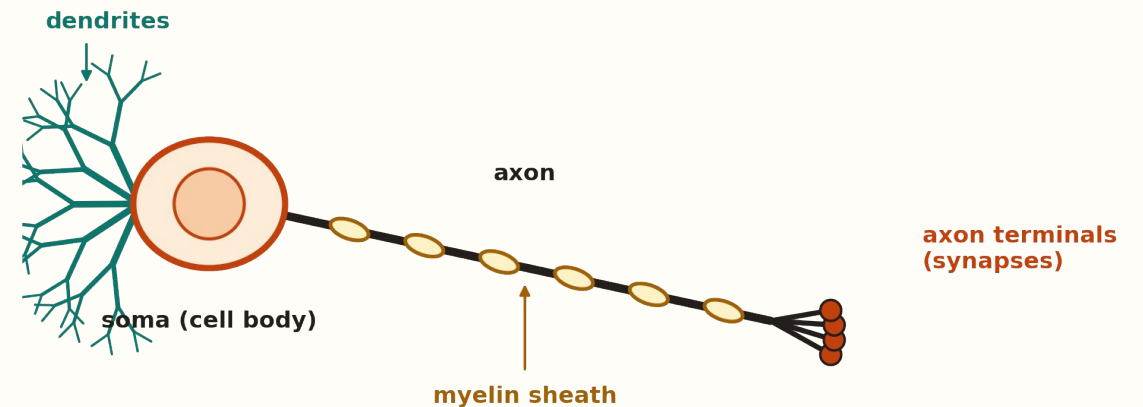
## Membrane potential

Variable is an unknown quantity that evolves over time and depends on other quantities. Parameter is a fixed quantity that shape the behavior of a system but does not influence the core evolution of a variable. Generally, It is chosen before you ever press “run” on a simulation.

- Variable : Membrane potential. It changes at every moment
- Parameters : resting potential, threshold : fixed numbers

Dependent variable : the value we want to find

Independent variable : the value that drives the system's evolution



*The anatomy doesn't change turn to turn but the electrical state living on top of it does.*

# Variables, parameters

In this equation, exactly one symbol is a variable, and two are parameters.

$$dV/dt = -(V - V_{rest}) / \tau$$

***V***

## The variable

Changes at every instant, this is what the equation solves for over time.

***$\tau$***

## A parameter

The membrane time constant. Fixed before the simulation starts; shapes HOW FAST V responds.

***V<sub>rest</sub>***

## A parameter

The resting potential. Also fixed in advance; sets WHERE V settles.

## Try this:

next time you see an unfamiliar equation in a paper, before reading the prose, just label every symbol as “variable” or “parameter.” It demystifies the equation immediately.

# Function

A function is the rule that connects everything:  $V(t)$  is a function of time; a firing-rate nonlinearity  $f(V)$  is a function of voltage. In a differential equation, we describe the rate of change of a variable as a function of the variable itself, other variables, and the parameters.

$dV/dt = F(V, t; \text{parameters})$  — this abstract form is what every equation in this lecture will turn out to be a special case of.

$$dV/dt = -(V - V_{rest}) / \tau$$

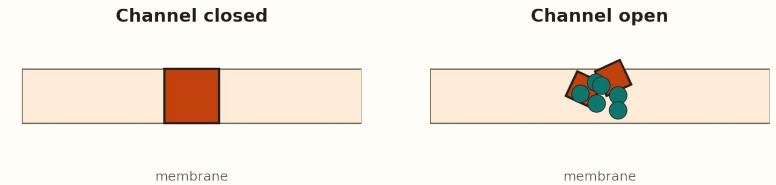
$$F_{t, V_{rest}}(V) = F(V, V_{rest}, t) = -(V - V_{rest}) / \tau$$

**The function**

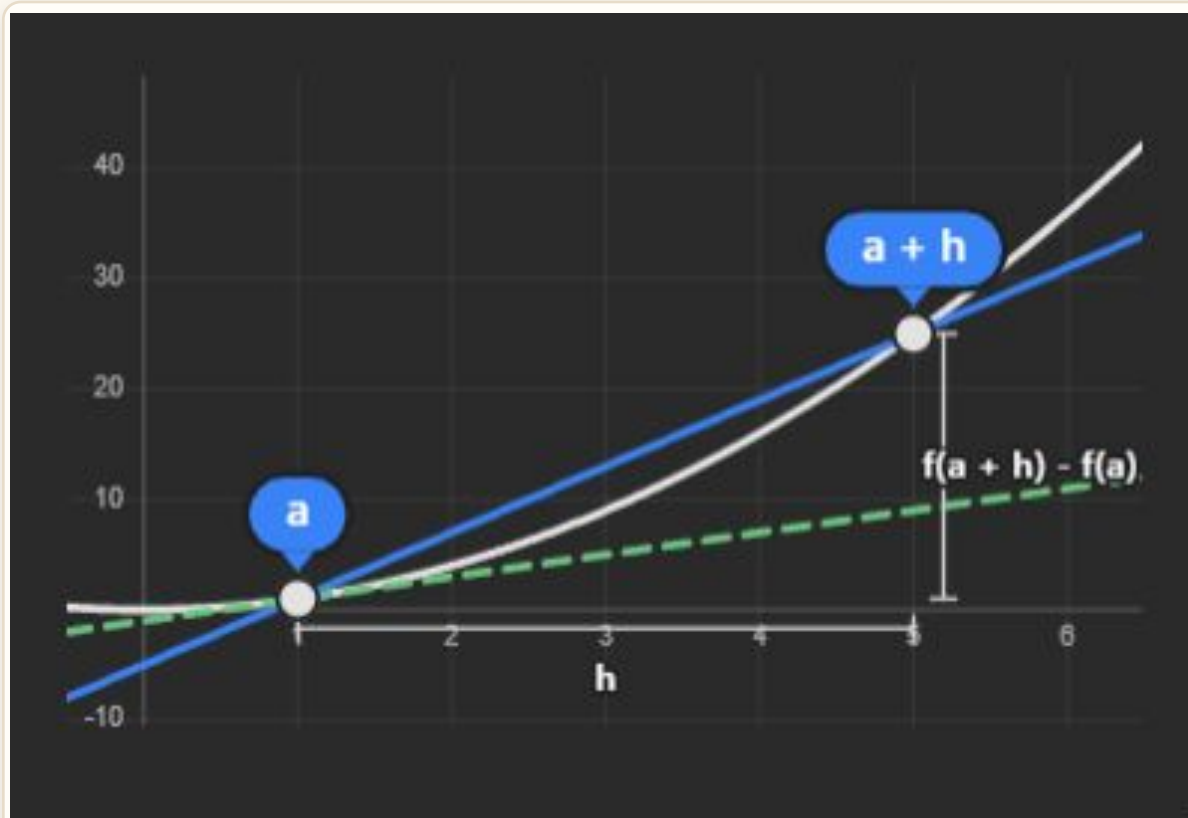
# 3

## Rate of change: slope and derivative

*A speedometer for anything, not just cars.*



# Rate of change: slope and derivative



Secant (average slope, blue dotted) versus tangent (instantaneous slope, orange dashed) on the same relaxation curve.

## Reading this figure

The two blue dots and the line between them are the secant, an average over an interval. As that interval shrinks to a point, the line's slope converges to the tangent's slope: the derivative, exactly at the orange dot.

- green dashed line : derivative
- blue line : slope

# Rate of change: slope and derivative

In the previous slide , watch how the secant line (blue) becomes the tangent line (orange) as the two points move together.

$$\text{average slope (secant)} \rightarrow \text{instantaneous slope (tangent)} = dV/dt$$

$$\Delta V/\Delta t$$

## Average slope

The secant line's slope, over some finite time interval.

$$\lim \Delta t \rightarrow 0$$

## The limiting process

Shrink the interval toward zero and the two points merge into one.

$$dV/dt$$

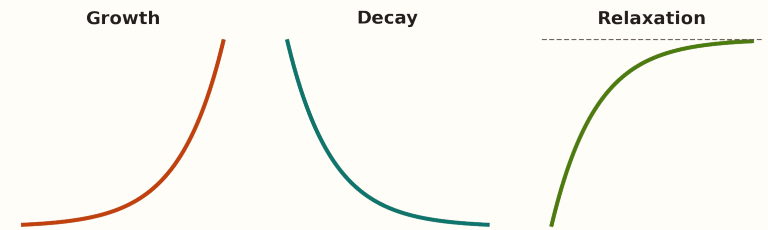
## Instantaneous slope

The tangent line's slope, at a single point in time.

# 4

## First-order equations: growth, decay, relaxation

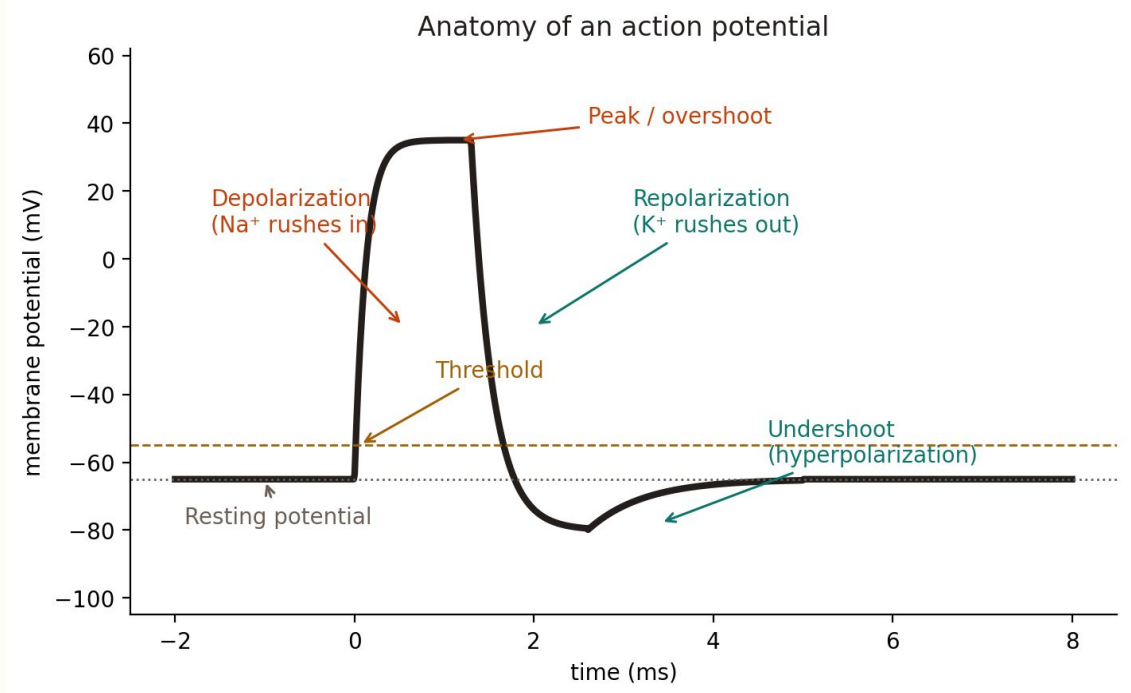
*Three patterns that cover almost everything you'll meet.*



# Action potential

*The rising and falling phases of a spike are both here.*

The sharp upstroke is, for a brief moment, almost like unchecked growth. The recovery afterward is relaxation decaying back toward a target. Two of our three canonical patterns, both visible in one biological event.



*Depolarization looks like runaway growth; the return to rest looks like relaxation. Same trace, two of our three patterns.*

# First-order equations: growth, decay, relaxation

Three formulas, one family — decay is relaxation with  $V_{eq}=0$ , and growth is decay with the sign flipped.

$$dx/dt = k \cdot x \text{ (growth)} \quad dx/dt = -k \cdot x \text{ (decay)} \quad dV/dt = -(V - V_{eq})/\tau \text{ (relaxation)}$$

## *Growth*

$$dx/dt = k \cdot x$$

## *Decay*

$$dx/dt = -k \cdot x$$

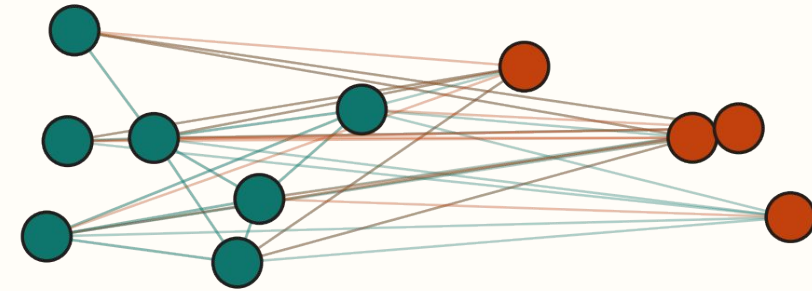
## *Relaxation*

$$dV/dt = -(V - V_{eq})/\tau$$

# Why it matters

*A resting network is a balance, not an absence of activity.*

The network on the right has real activity flowing through it. Excitatory and Inhibitory connections constantly pushing and pulling. “Stability” doesn't mean nothing is happening; it means the pushes and pulls balance out at some point, and small nudges away from that balance get corrected.



● excitatory population (E) ● inhibitory population (I)

*Excitatory (teal) and inhibitory (terracotta) connections in balance*

# Equilibrium points and stability

*“Finding an equilibrium tells you where a system CAN sit still. Stability tells you what happens when it's nudged.”*

An equilibrium point (fixed point, steady state) is a value where the rate of change is exactly zero. If the system starts exactly there, it stays there forever. For our anchor equation, setting  $dV/dt=0$  immediately gives  $V^*=V_{\text{rest}}$ : the resting potential IS the equilibrium, by construction. But finding an equilibrium only tells you where the system CAN sit still. Stability asks what happens when it's nudged away: a STABLE equilibrium pulls perturbations back; an UNSTABLE one pushes them further away.

## The rule for linear equations:

for  $dx/dt = a \cdot (x - x^*)$ , the equilibrium is stable if  $a < 0$ , and unstable if  $a > 0$ . The sign of that one coefficient is the entire story.

# Equilibrium points and stability

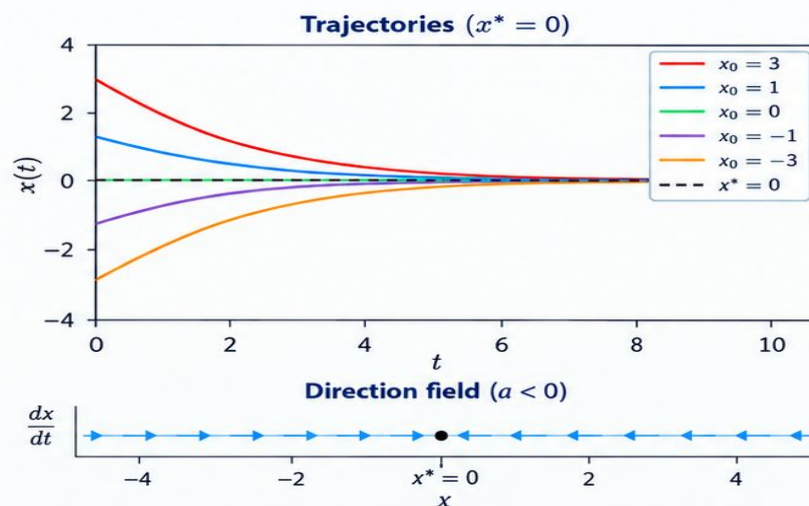
## Stable equilibrium ( $a < 0$ )

$$\frac{dx}{dt} = a(x - x^*) \quad \text{with } a = -0.5 \quad (\text{example})$$

Solution:

$$x(t) = x^* + (x_0 - x^*)e^{at}$$

Since  $a < 0$ ,  $e^{at} \rightarrow 0$  as  $t \rightarrow \infty$ .  
All trajectories approach  $x^*$ .



### Key takeaways:

- The equilibrium is  $x^*$  because  $f(x^*) = 0$ .
- The solution  $x(t) = x^* + (x_0 - x^*)e^{at}$  shows what happens over time.
- If  $a < 0$ : deviations shrink  $\rightarrow$  stable equilibrium.
- If  $a > 0$ : deviations grow  $\rightarrow$  unstable equilibrium.

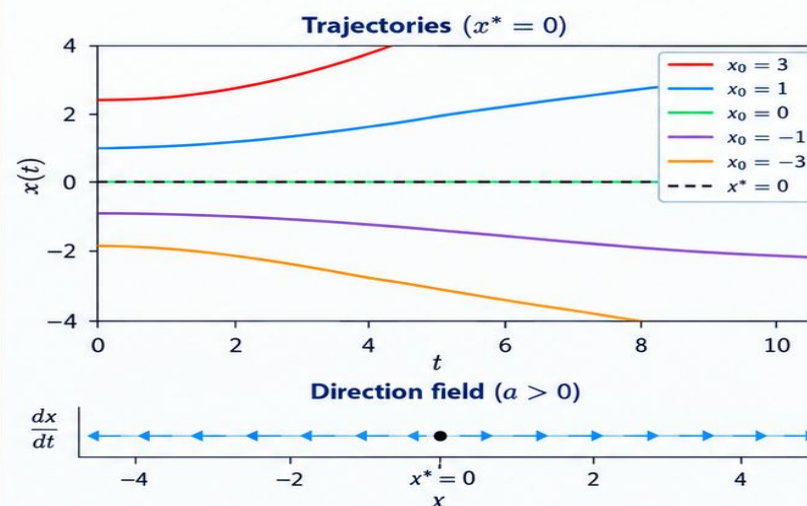
## Unstable equilibrium ( $a > 0$ )

$$\frac{dx}{dt} = a(x - x^*) \quad \text{with } a = 0.5 \quad (\text{example})$$

Solution:

$$x(t) = x^* + (x_0 - x^*)e^{at}$$

Since  $a > 0$ ,  $e^{at} \rightarrow \infty$  as  $t \rightarrow \infty$ .  
Trajectories move away from  $x^*$ .



### In short:

- **Stable:** if you start near  $x^*$ , you stay near  $x^*$ .
- **Unstable:** if you start near  $x^*$ , you move away.

# Test yourself before moving on

*For  $dx/dt = 0.3 \cdot (x-5)$ , is  $x=5$  stable or unstable?*

Q1

Unstable — the coefficient, 0.3, is positive.

*What two questions let you characterize a system's long-run behaviour without solving it explicitly?*

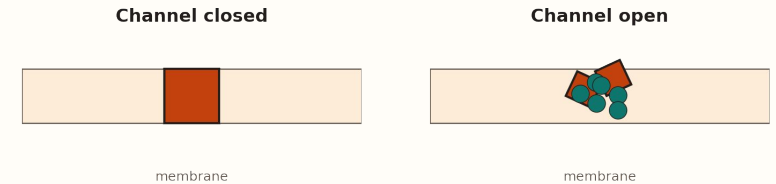
Q2

Where is  $dx/dt=0$ ? And does the system move toward or away from that point?

# 5

## Leaky dynamics and the resting state

*Why biology keeps reaching for the same safe, self-correcting pattern.*



# Leaky dynamics and the resting state

This is the same relaxation equation from Section 5, now examined for WHY it behaves the way it does.

$$dx/dt = -(x - x_{eq}) / \tau \quad \text{self-correcting, history-forgetting, single time-scale}$$

## *Self-correcting*

### **No controller needed**

Perturbations are automatically opposed by the equation's own structure.

## *History-forgetting*

### **After $\sim 5\tau$**

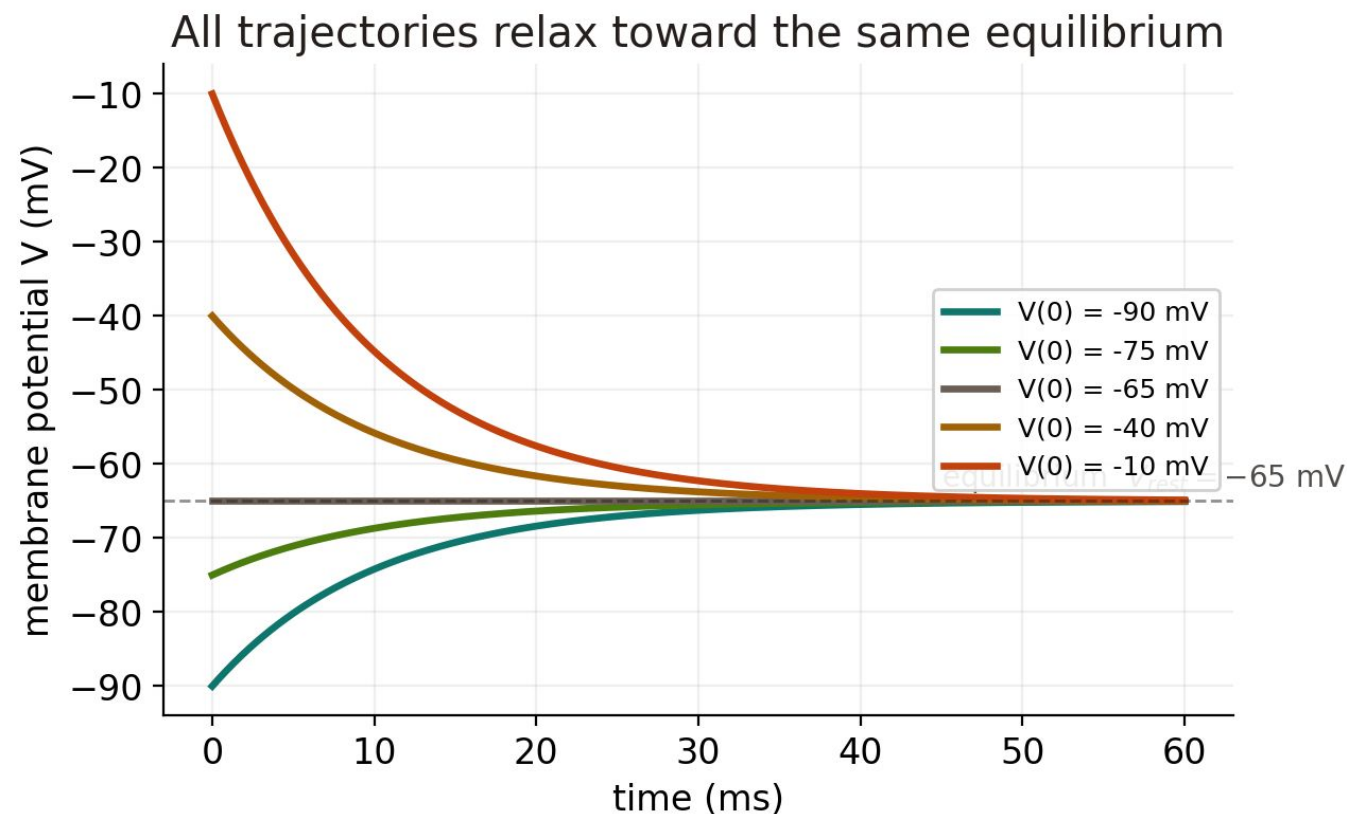
The system looks the same regardless of where it started.

## *Single time-scale*

### **Governed by $\tau$ alone**

No oscillation, no overshoot , just a smooth approach.

# Leaky dynamics and the resting state



Five different starting voltages, all relaxing toward the same equilibrium — larger displacement produces a steeper initial slope, but the same long-run destination.

## What this figure proves

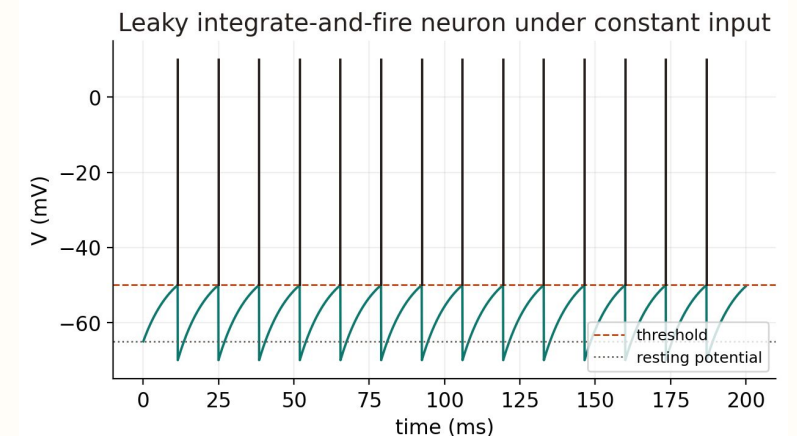
Every curve, regardless of starting point, converges on  $-65$  mV. That convergence, independent of history, is the signature of a stable equilibrium made visible in a simulation rather than an abstract phase-line diagram.

$$\frac{dV}{dt} = - \underbrace{\frac{V - V_{\text{rest}}}{\tau_m}}_{\text{leak}} + \underbrace{\frac{R_m I(t)}{\tau_m}}_{\text{input}}$$

# 6

## Membrane potential models: leaky integrate-and-fire

*Smooth calculus, punctuated by one discrete rule.*



# Membrane potential models: leaky integrate-and-fire

Read the differential part first — it's entirely familiar — then treat the spike rule as a separate, second layer on top.

$$\tau \cdot dV/dt = -(V - V_{rest}) + R \cdot I(t) \quad \text{then: if } V \geq V_{th}: \text{spike, } V \leftarrow V_{reset}$$

$$-(V - V_{rest})$$

## The familiar leak

Exactly Section 5's restoring push toward rest.

$$R \cdot I(t)$$

## The new term

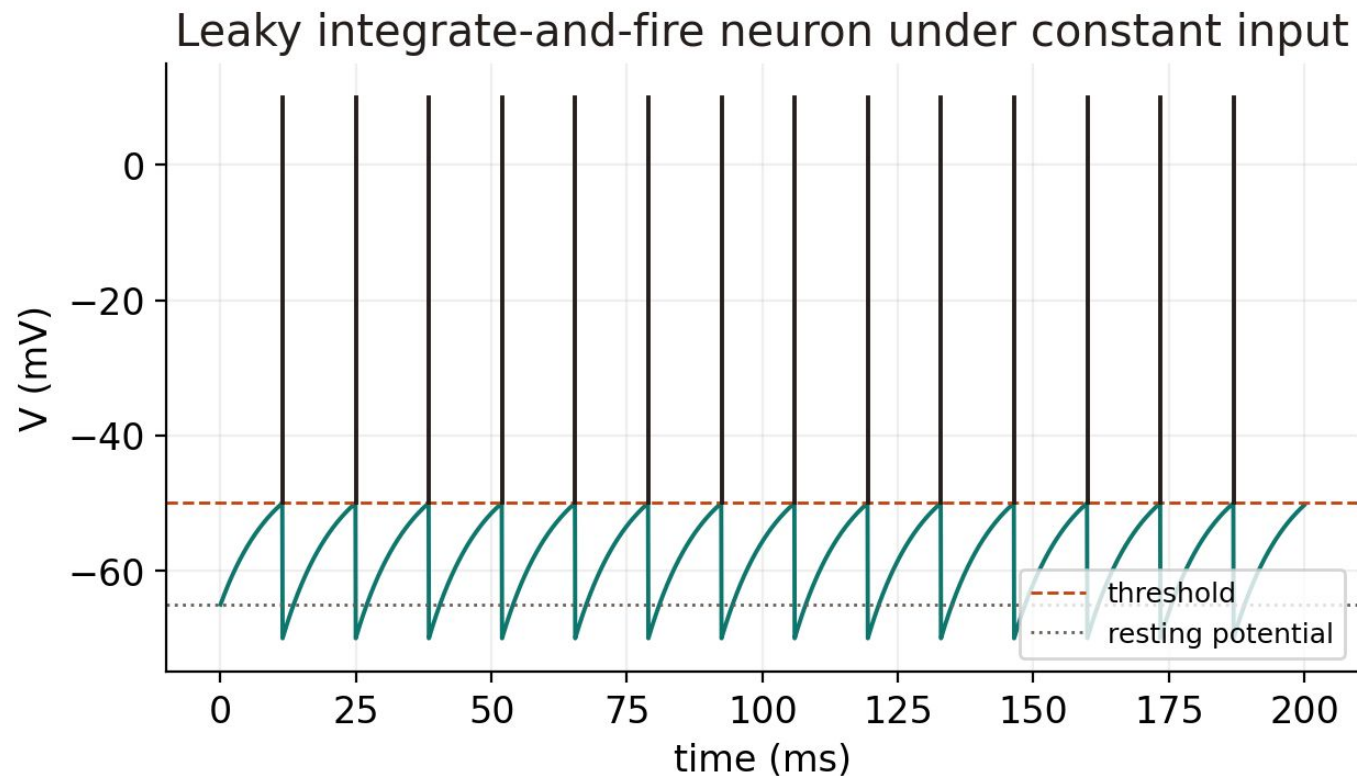
Injected or synaptic current, pushing V away from rest.

$$V \geq V_{th} \rightarrow \text{reset}$$

## The discrete rule

Not part of the differential equation, a separate condition checked at each step.

# Membrane potential models: leaky integrate-and-fire

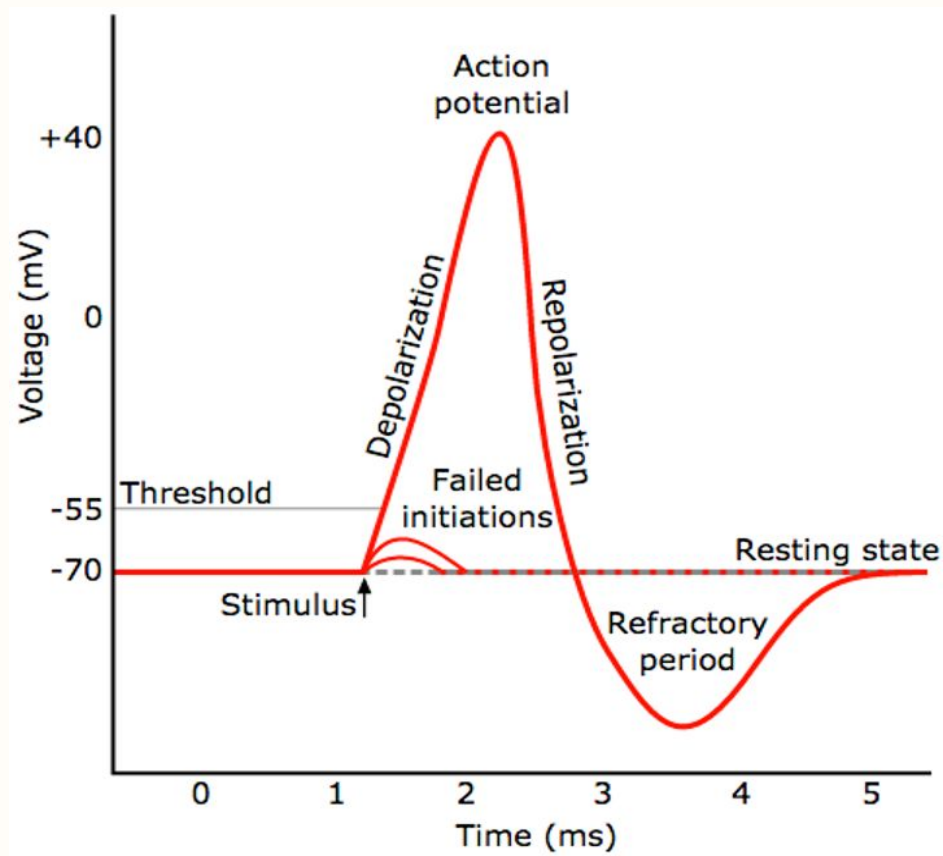


## Key idea

With constant input,  $V$  is driven toward  $V_{eq} = V_{rest} + R \cdot I$ . Whether the neuron ever fires depends entirely on whether this target sits above or below threshold — Section 6's equilibrium concept, made concrete.

*A leaky integrate-and-fire neuron under constant input current. Between spikes,  $V$  follows the leaky relaxation curve upward; each threshold crossing fires and resets.*

# What a single spike actually looks like



source : Bioserver - Understanding the spikes of neurons  
The LIF model replaces this entire shape with an instant vertical reset

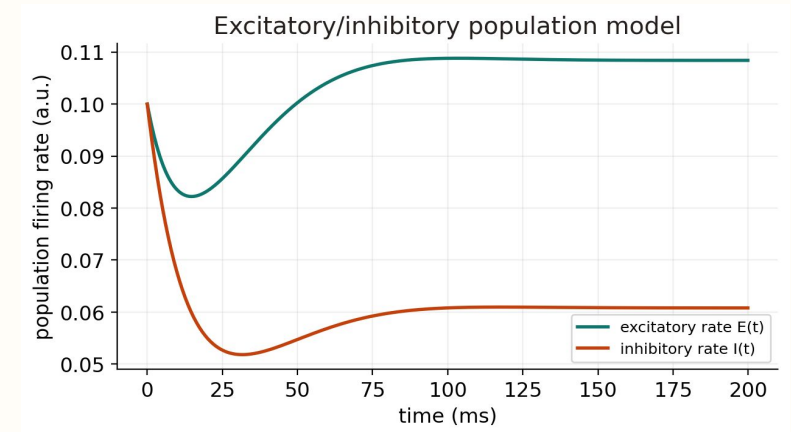
## Why the simplification is OK

LIF throws away the shape of the spike itself (depolarization, overshoot, repolarization) and keeps only the TIMING of threshold crossings. For most circuit-level questions, firing rates, spike-timing patterns, that's exactly the information that matters.

# 7

## Neural population models

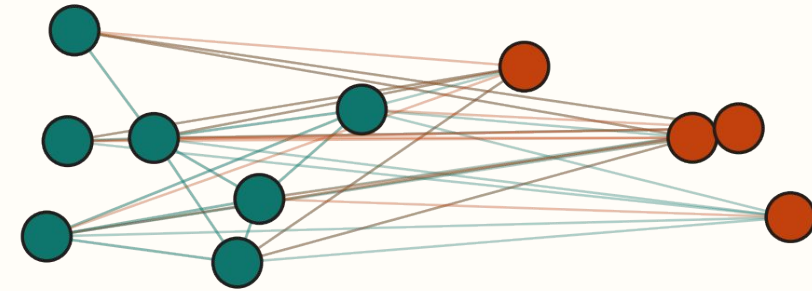
*The same leak-vs-drive logic, now with two variables instead of one.*



# Why it matters

## *Thousands of synapses, summarized in four numbers.*

Every connection in the network illustration is a synapse like the one from our earlier primer — a presynaptic release event changing a postsynaptic current. A population model doesn't track each one individually; it summarizes millions of them into a handful of coupling weights ( $w_{EE}$ ,  $w_{EI}$ ,  $w_{IE}$ ,  $w_{II}$ ) and lets two variables,  $E$  and  $I$ , describe the whole circuit's average behaviour.



● **excitatory population (E)** ● **inhibitory population (I)**

*An excitatory/inhibitory network — the biological substrate that the Wilson-Cowan equations compress into two coupled variables.*

# Neural population models

Two coupled equations, each with the exact same leak-vs-drive structure as every equation so far.

$$\tau_E \cdot dE/dt = -E + f(w_{EE} \cdot E - w_{EI} \cdot I + P_E) \quad \tau_I \cdot dI/dt = -I + f(w_{IE} \cdot E - w_{II} \cdot I + P_I)$$

$-E, -I$

## The leak terms

Each population's rate is always pulled back toward zero on its own.

$f(\cdot)$

## The nonlinearity

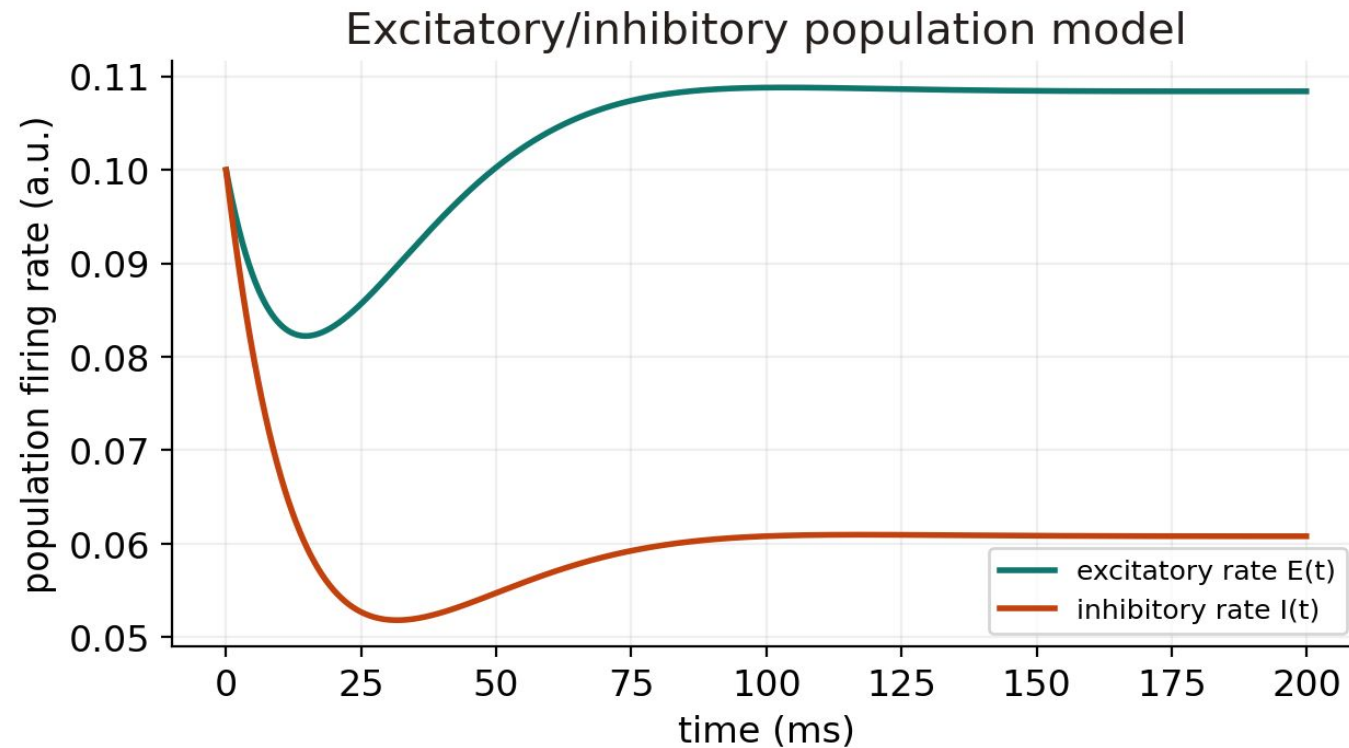
A sigmoid, ensuring rates saturate and never go negative.

$w_{EE}, w_{EI}, w_{IE}, w_{II}$

## The coupling weights

Summarize how strongly each population drives the other — and itself.

# Neural population models

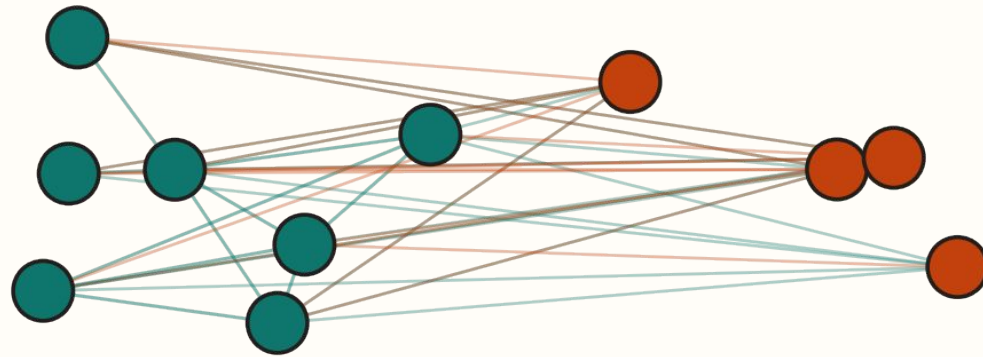


*Excitatory (teal) and inhibitory (terracotta) population rates, evolving from the same simple starting point, settling into a stable E/I balance.*

## Why this matters

Two interacting variables have a much richer space of dynamics than one ; including multiple stable states (relevant to working memory) and sustained oscillations, which a single leaky variable structurally cannot produce.

# From network to two numbers



● **excitatory population (E)** ● **inhibitory population (I)**

*Every teal-to-terracotta and terracotta-to-teal connection above is folded into just two coupling weights,  $w_{EI}$  and  $w_{IE}$  — a massive simplification that still captures the E/I balance.*

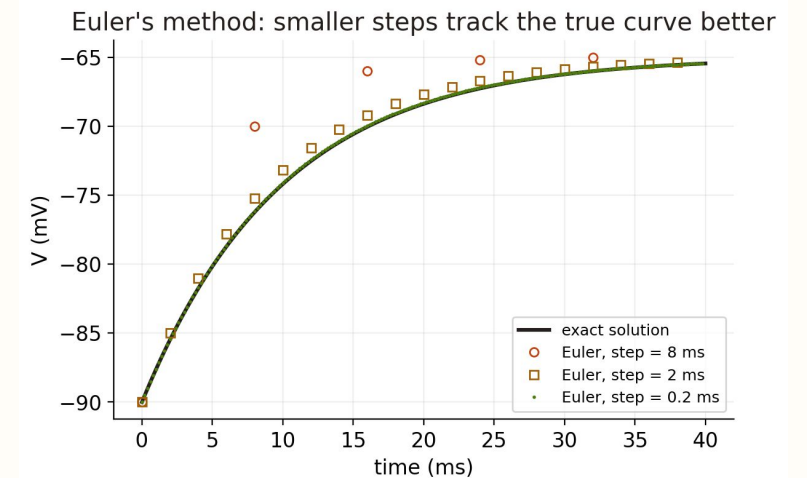
## Why this simplification works

For many circuit-level questions, individual connections don't matter. Only their AGGREGATE effect does. Two coupling numbers per direction is often enough to reproduce a network's qualitative behaviour.

# 8

## Numerical approximation: simulating step by step

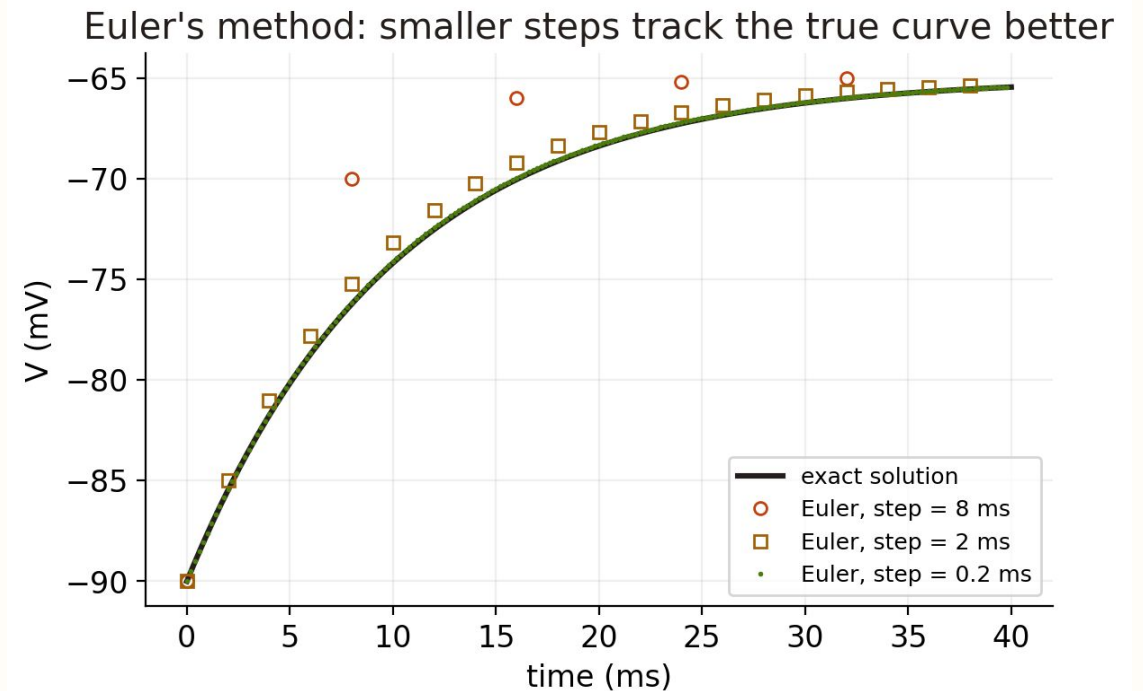
*Every simulation you've run today was already doing this.*



# Why it matters

*You've been using Euler's method since Section 8 without naming it.*

Look back at the LIF and Wilson-Cowan code: the line  $V[i] = V[i-1] + dt \cdot dV$  is the entire technique this section formalizes. For only a handful of especially simple equations, like plain leaky decay, can we write an exact formula for  $V(t)$ . **For almost everything realistic, we approximate numerically** instead.



*The same relaxation equation solved with Euler's method at three step sizes, against the exact analytic solution.*

# Numerical approximation: simulating step by step

This one line, applied repeatedly in a loop.

$$x(t + \Delta t) \approx x(t) + \Delta t \cdot [dx/dt \text{ at time } t]$$

## *Compute the slope*

### Step 1

Evaluate  $dx/dt$  at the current state, using the model's rule.

## *Take a small step*

### Step 2

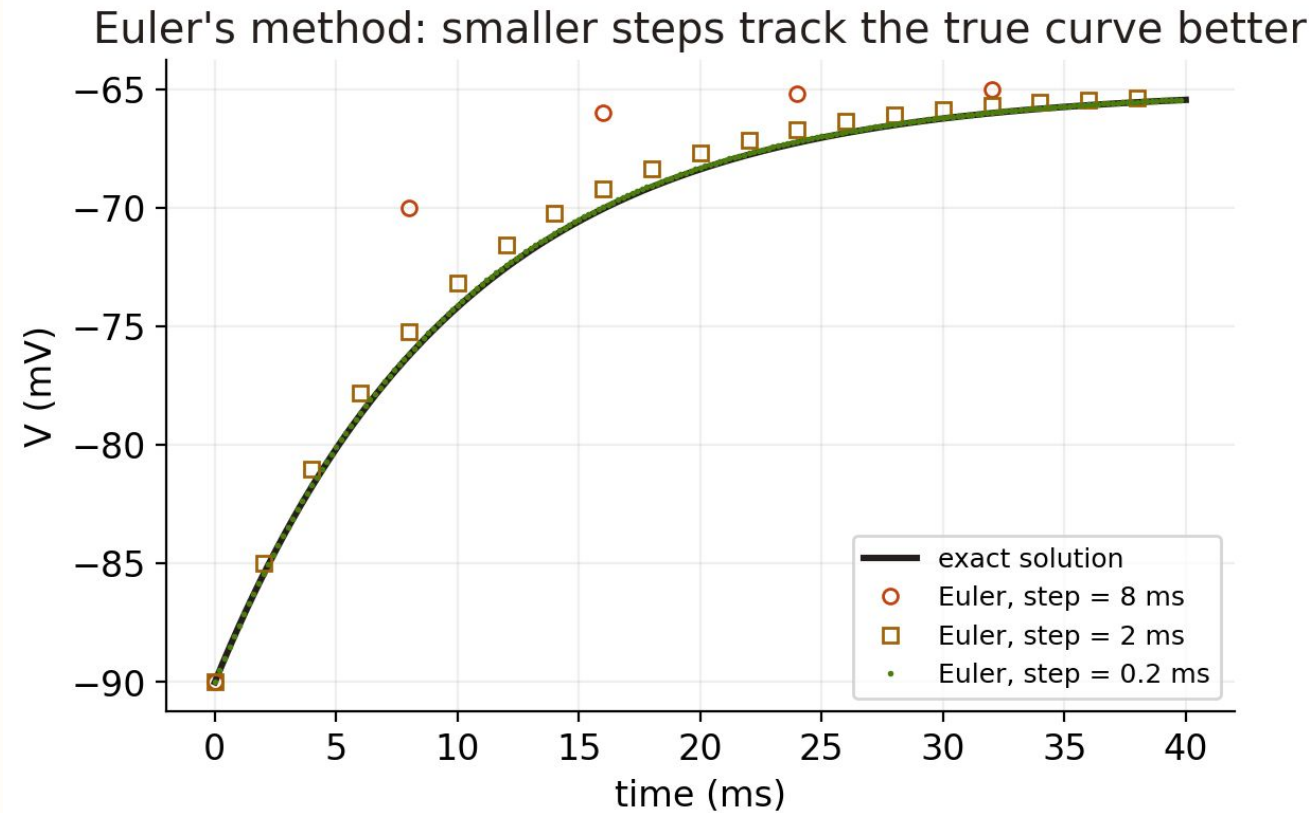
Move forward by  $\Delta t$ , assuming the slope stays constant over that step.

## *Repeat*

### Step 3

Re-evaluate the slope at the new point, and continue.

# Numerical approximation: simulating step by step

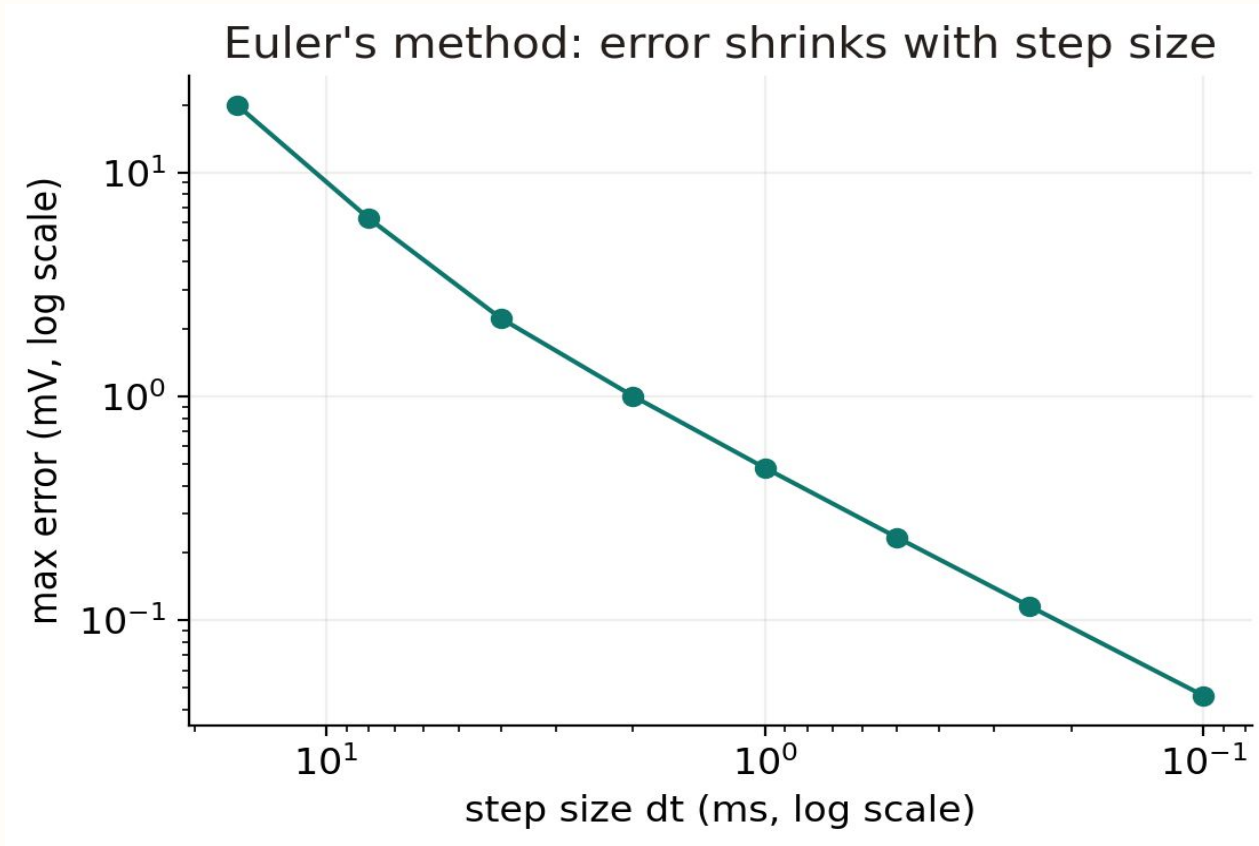


*The same relaxation equation solved with Euler's method at three step sizes, against the exact analytic solution.*

## Reading this figure

The coarsest step (red) visibly lags behind the true curve. The finest step (green) is nearly indistinguishable from the exact solution. Everything in between is the accuracy/speed trade-off, made visible.

# Does it actually converge? Check the error



Maximum error against the exact solution, at several step sizes, on log-log axes.

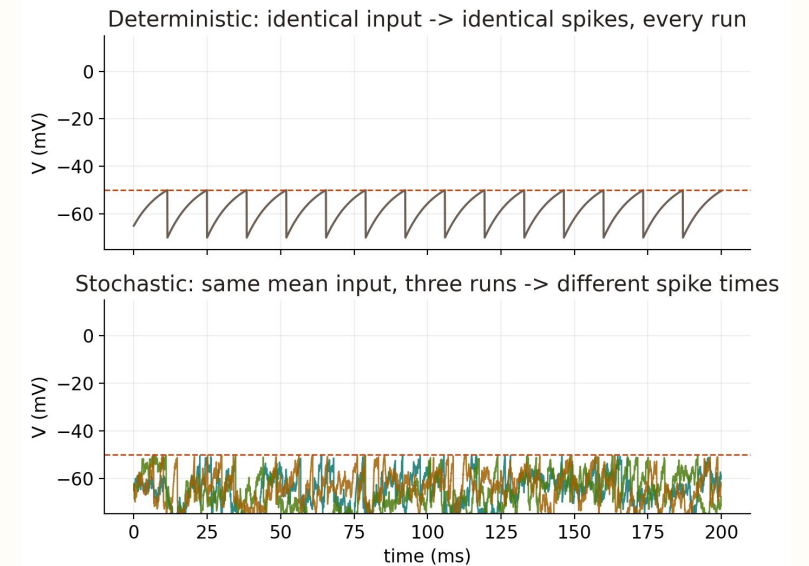
## Reading a log-log plot

On log-log axes, error vs. step size falls on a roughly straight line, the visual signature of convergence. This is stronger evidence than “the curve looks smoother”, it's quantitative proof that shrinking  $dt$  approaches the truth.

# 9

## Deterministic vs. stochastic models

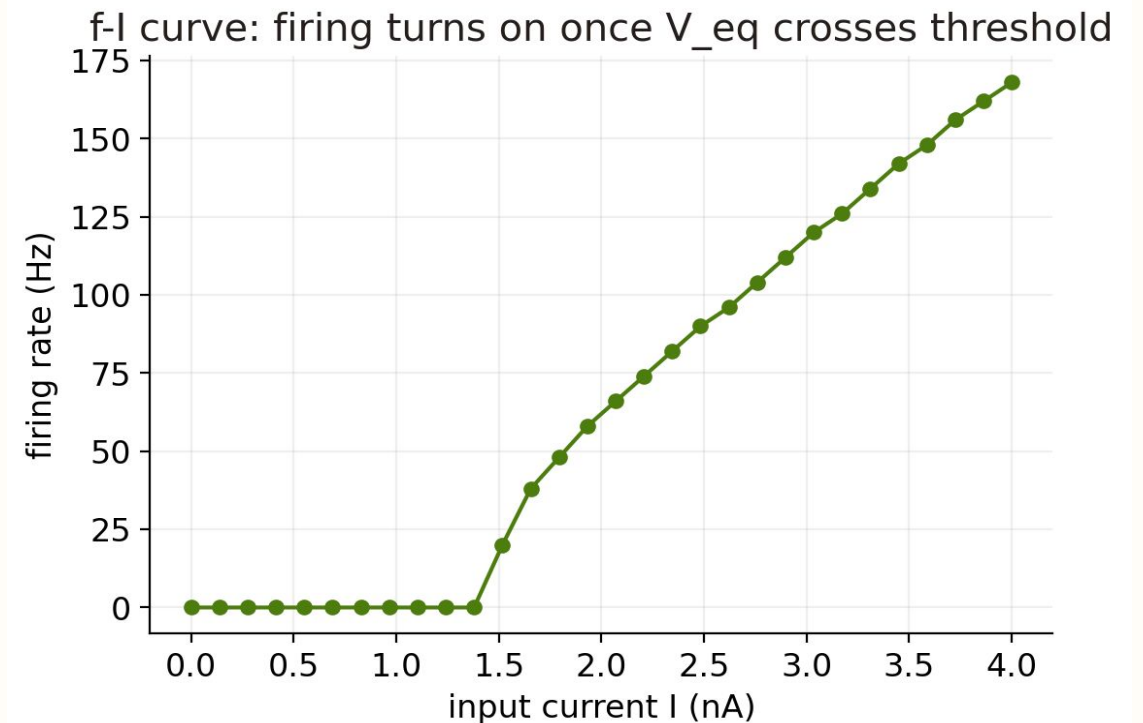
*Real neurons don't give the same answer twice.*



# Why it matters

*The  $f$ - $I$  curve you saw was already an average.*

Real neurons are noisier: synaptic release is probabilistic, channels open and close stochastically, and background input from thousands of other neurons looks effectively random. That smooth curve is a deterministic MEAN behaviour, hiding real trial-to-trial variability underneath.



*A clean  $f$ - $I$  relationship — but every real neuron deviates from this mean on any given trial.*

# Deterministic vs. stochastic models

One extra additive term, with one crucial scaling detail, converts any deterministic model into its stochastic counterpart.

$$dV/dt = -(V - V_{rest})/\tau + R \cdot I/\tau + \text{noise}(t) \quad \text{noise scaled by } \sqrt{\Delta t}$$

## ***Deterministic core***

### **Unchanged**

The same leak-vs-drive equation from every previous section.

## ***noise(t)***

### **The new term**

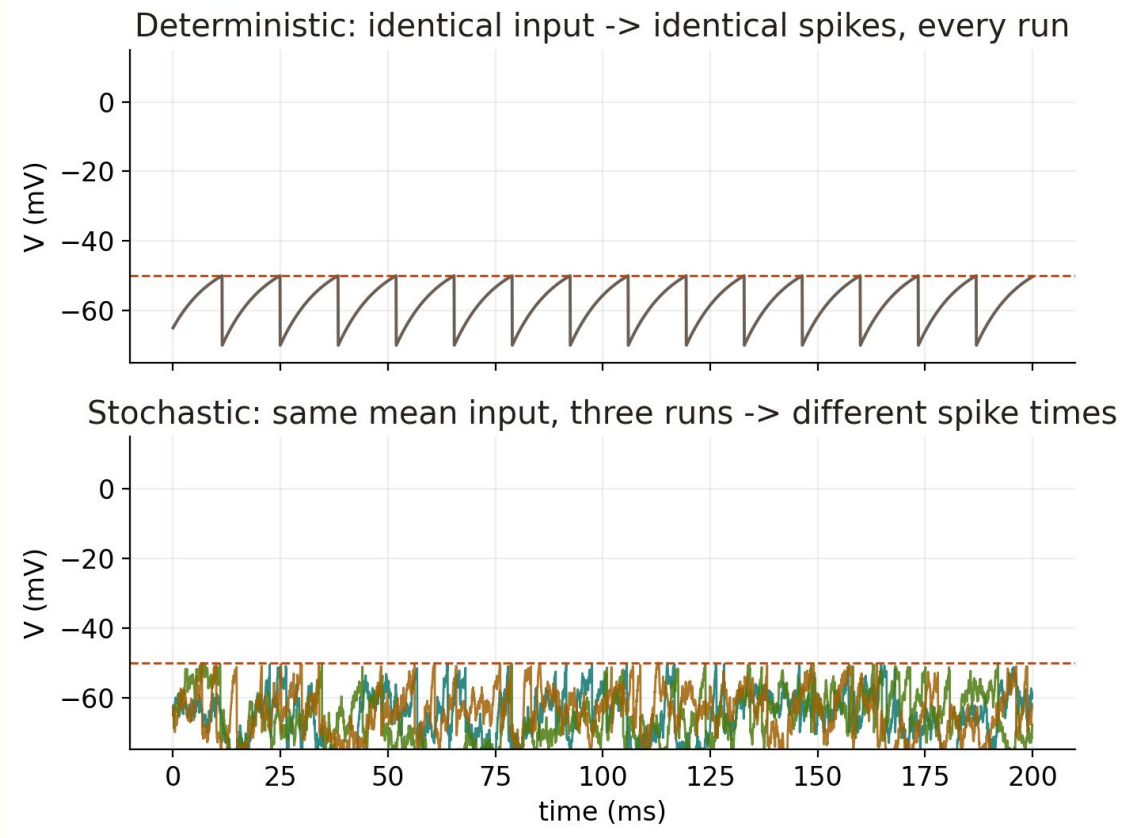
A random Gaussian kick, drawn fresh at every simulation step.

## ***$\times \sqrt{\Delta t}$***

### **The critical scaling**

Keeps total injected randomness independent of your chosen step size.

# Deterministic vs. stochastic models



Top: deterministic LIF — identical spikes, every run. Bottom: the same neuron with noise — three runs, same mean input, three different spike-time patterns.

## Why this distinction matters

Deterministic models are easiest to analyze and best for typical, mean behaviour. Stochastic models are necessary when variability itself is the phenomenon : spike-timing reliability, or noise-driven switching between states in a bistable circuit.

# One mental model, reused at every scale

1

## Identify variables & parameters

What changes over time, and what's fixed?

2

## Write the rate of change

$dx/dt = F(x,t)$  — read it as a sentence about competing pushes.

3

## Find equilibria & check stability

Set  $dx/dt = 0$ ; ask if nearby points are pushed toward or away.

4

## Recognize leaky/relaxation form

$dx/dt = -(x - x_{eq})/\tau$  — it is everywhere in neuroscience.

5

## Simulate when no formula exists

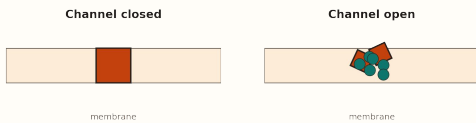
Step forward with Euler's method; shrink the step to check convergence.

6

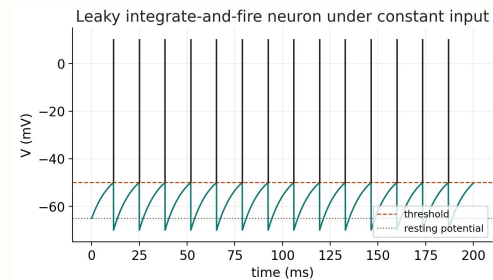
## Choose deterministic or stochastic

Mean behaviour, or the variability itself?

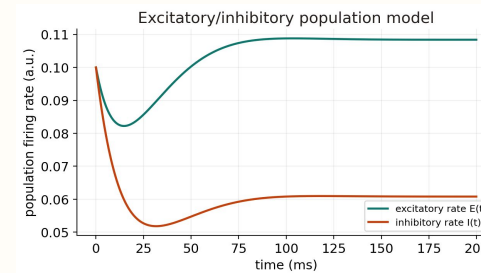
# The same equation, four scales — revisited



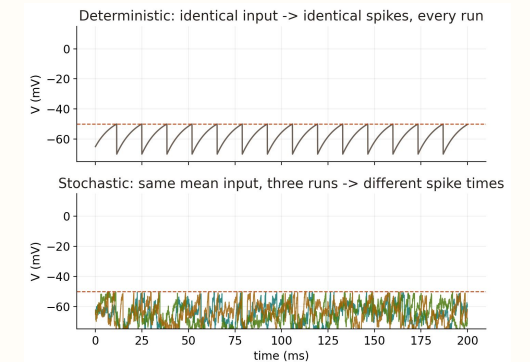
*A channel opening/closing*



*A single neuron's voltage*



*A population's firing rate*



*Variability across trials*

# Where to go from here

1

## Hodgkin-Huxley models

Replace the single leak current with explicit, voltage-gated  $\text{Na}^+$  and  $\text{K}^+$  conductances — the biophysical origin of the action potential shape from Section 1.

2

## Larger spiking networks

Thousands of LIF neurons, connected the way our Section 9 network illustration suggested — emergent dynamics from many simple units.

3

## Neural mass models

Wilson-Cowan-style equations, scaled up to interpret EEG and fMRI signals at the whole-brain level.

4

## Reinforcement learning models

Differential equations describing dopaminergic signalling and value updating — same toolkit, a different biological variable.

# Questions worth sitting with

1

Where else in your own data might you look for a leaky / relaxation pattern?

2

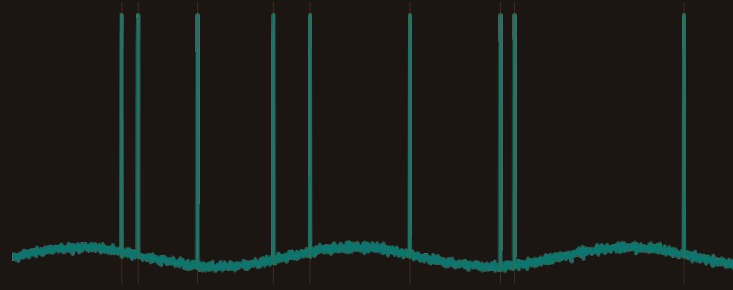
For a system you study, what is the equilibrium — and is it stable?

3

Would your question be better served by the mean (deterministic) behaviour, or by the variability itself (stochastic)?

4

What would it take to extend a single-neuron model to a population model of your circuit of interest?



# Thank you

*Questions & discussion*